Applications: Growth and Decay

Growth models

1 A city's population is modeled by $P(t) = 12000(1.035)^t$, where t is in years.

a State the initial population and the annual growth rate.

b Find the population after 10 years, to the nearest person.

2 An account earns 4% annual interest compounded continuously. How long does it take an investment to double? Give an exact expression and a decimal to two places.

Decay models

3 Carbon-14 has a half-life of 5730 years: $A(t) = A_0\left(\frac{1}{2}\right)^{t/5730}$.

a What fraction of the original carbon-14 remains after 2000 years? Answer to three decimal places.

b A sample retains 30% of its carbon-14. Estimate its age to the nearest hundred years.

4 A 200 g sample of a radioactive substance decays to 50 g in 12 hours, following $A(t) = 200e^{kt}$.

a Find the decay constant k (4 d.p.).

b Find the half-life of the substance.

5 Newton's law of cooling: coffee at $90°$C cools in a $20°$C room following $T(t) = 20 + 70e^{-0.1t}$ (t in minutes). Find the temperature after 10 minutes, to one decimal place.